## Guess the Output Quiz (20 Questions)

Read each program carefully and write down the output. If the code will not compile, write "Compile-time error". Answers are given at the end.

**Q1. What is the output?**

```
static void show() {
    System.out.println("Hi");
}
public static void main(String[] a) {
    show();
    show();
}
```

Answer: _____

**Q2. What is the output?**

```
static void greet(String n) {
    System.out.println("Hello " + n);
}
public static void main(String[] a) {
    greet("Ravi");
}
```

Answer: _____

**Q3. What is the output?**

```
static int five() {
    return 5;
}
public static void main(String[] a) {
    System.out.println(five() + 2);
}
```

Answer: _____

**Q4. What is the output?**

```
static int add(int a, int b) {
    return a + b;
}
public static void main(String[] a) {
    System.out.println(add(3, 4) * 2);
}
```

Answer: _____

**Q5. What is the output?**

```
static void print(int x) {
    x = x + 10;
    System.out.println(x);
}
public static void main(String[] a) {
    int n = 5;
    print(n);
    System.out.println(n);
}
```

Answer: _____

**Q6. What is the output?**

```
static int square(int n) {
    return n * n;
}
public static void main(String[] a) {
    System.out.println(square(square(2)));
}
```

Answer: _____

**Q7. What is the output?**

```
static int test() {
    return 1;
    System.out.println("After");
}
public static void main(String[] a) {
    System.out.println(test());
}
```

Answer: _____

**Q8. What is the output?**

```
static String say() {
    return "Java";
}
public static void main(String[] a) {
    System.out.println(say().length());
}
```

Answer: _____

**Q9. What is the output?**

```
static void m(int a) {
    System.out.println("int");
}
static void m(double a) {
    System.out.println("double");
}
public static void main(String[] a) {
    m(5);
    m(5.0);
}
```

Answer: _____

**Q10. What is the output?**

```
static void m(long a) {
    System.out.println("long");
}
static void m(double a) {
    System.out.println("double");
}
public static void main(String[] a) {
    m(5);
}
```

Answer: _____

**Q11. What is the output?**

```
static boolean isEven(int n) {
    return n % 2 == 0;
```

```
}
public static void main(String[] a) {
    System.out.println(isEven(7));
}
```

Answer: _____

**Q12. What is the output?**

```
static int fact(int n) {
    if (n == 0) return 1;
    return n * fact(n - 1);
}
public static void main(String[] a) {
    System.out.println(fact(4));
}
```

Answer: _____

**Q13. What is the output?**

```
static void change(int[] arr) {
    arr[0] = 99;
}
public static void main(String[] a) {
    int[] x = {1, 2, 3};
    change(x);
    System.out.println(x[0]);
}
```

Answer: _____

**Q14. What is the output?**

```
static void change(String s) {
    s = s + " World";
}
public static void main(String[] a) {
    String t = "Hello";
    change(t);
    System.out.println(t);
}
```

Answer: _____

**Q15. What is the output?**

```
static int sum(int... n) {
    int s = 0;
    for (int x : n) s += x;
    return s;
}
public static void main(String[] a) {
    System.out.println(sum(1, 2, 3, 4));
}
```

Answer: _____

**Q16. What is the output?**

```
static double half(int n) {
    return n / 2;
}
public static void main(String[] a) {
    System.out.println(half(5));
}
```

Answer: _____

**Q17. What is the output?**

```
static int count = 0;
static void inc() {
    count++;
}
public static void main(String[] a) {
    inc();
    inc();
    inc();
    System.out.println(count);
}
```

Answer: _____

**Q18. What is the output?**

```
static int f(int a, int b) {
    return a - b;
}
public static void main(String[] x) {
    System.out.println(f(10, 4) + " " + f(4, 10));
}
```

Answer: _____

**Q19. What is the output?**

```
static char grade(int m) {
    if (m >= 50) return 'P';
    return 'F';
}
public static void main(String[] a) {
    System.out.println(grade(49) + "" + grade(50));
}
```

Answer: _____

**Q20. What is the output?**

```
static void show() {
    System.out.print("A");
}
static int get() {
    show();
    return 7;
}
public static void main(String[] a) {
    System.out.println(get());
}
```

Answer: _____

## Answer Key

| Q | Output | Explanation |
|---|---|---|
| **1** | `Hi`<br>`Hi` | show() is called twice, so it prints twice. |
| **2** | `Hello Ravi` | The argument "Ravi" is joined to "Hello ". |
| **3** | `7` | five() returns 5, then 5 + 2 = 7. |
| **4** | `14` | add(3, 4) = 7, then 7 * 2 = 14. |
| **5** | `15`<br>`5` | Java passes primitives by value; the original n is unchanged. |
| **6** | `16` | square(2) = 4, then square(4) = 16. |
| **7** | `Compile-time error` | The statement after return is unreachable. |
| **8** | `4` | say() returns "Java", whose length is 4. |
| **9** | `int`<br>`double` | Overloading: Java picks the method that matches the argument type. |
| **10** | `long` | int is widened to the closest type, which is long. |
| **11** | `false` | 7 % 2 = 1, so the condition is false. |
| **12** | `24` | 4 * 3 * 2 * 1 = 24 (recursion). |
| **13** | `99` | Arrays are objects; the method changes the same array. |
| **14** | `Hello` | Strings are immutable and s is only a local copy of the reference. |
| **15** | `10` | Varargs collects all arguments into an array: 1+2+3+4 = 10. |
| **16** | `2.0` | Integer division gives 2, which is then widened to 2.0. |
| **17** | `3` | The static variable keeps its value between calls. |
| **18** | `6 -6` | Argument order matters: 10-4 = 6 and 4-10 = -6. |
| **19** | `FP` | grade(49) is F, grade(50) is P; they are joined as a String. |
| **20** | `A7` | show() prints A first (no newline), then 7 is printed. |